The Limits of Explanation

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Abstract

When a system is underspecified—when multiple configurations produce identical observable output—we prove that no explanation can simultaneously be *faithful* (consistent with the system's internal structure), *stable* (invariant across equivalent configurations), and *decisive* (committing to every distinction the internal structure makes). We derive this impossibility from first principles in eight scientific domains—underdetermined linear systems, genetic code degeneracy, gauge freedom, microstate multiplicity, syntactic ambiguity, the phase problem, the view update problem, and Markov equivalence—each requiring zero shared axioms. The framework is mechanically verified in the Lean 4 proof assistant (96 files, 424 theorems, 0 unproved goals), and empirically validated in seven domains with real data and negative controls. The constructive resolution—projecting onto the subspace invariant under the symmetry group of equivalent configurations—is provably Pareto-optimal and unifies existing domain-specific practices (gauge-invariant observables, the microcanonical ensemble, CPDAGs, minimum-norm solutions) as instances of a single G-invariant strategy that eight independent fields converged upon over more than a century. Cross-domain transfer experiments confirm the framework's predictive power: an ML ensemble method transferred to causal discovery reduces instability fourfold, and a character-theoretic bound derived from genetics predicts the rank ordering of protein sequence diversity (Spearman $\rho = 0.88$, $p = 0.004$), serving as a neutral-evolution upper bound that systematic deviations below quantify selective pressure. The framework provides a common formalization connecting Fisher's sufficiency criterion, the Rao-Blackwell theorem, and the EM algorithm to the same resolution structure, and gives a $1/n$ ceiling on neuron-level network interpretability under permutation symmetry—a bound that does not apply to circuit-level analysis, which is invariant under relabeling. For maximally incompatible systems—including binary rankings and DAG orientations—the impossibility strengthens to a bilemma: even faithful + stable is unachievable without enriching the explanation space.

Introduction

Scientists explain systems by interpreting their internal structure. A geneticist infers the DNA sequence that encodes a protein. A physicist chooses a gauge to describe an electromagnetic field. A statistician selects a causal graph from observational data. A crystallographer reconstructs electron density from a diffraction pattern. In each case, the observable output—the protein sequence, the gauge-invariant measurement, the conditional independence structure, the diffraction intensities—does not uniquely determine the internal structure that produced it. Multiple configurations yield identical observations. This situation is not a pathology of any single field; it is a structural feature of underspecified inference, present whenever the mapping from internal structure to observable output is many-to-one.

The consequences are practical and far-reaching. In clinical genomics, the inability to uniquely determine a coding sequence from an amino acid sequence limits the precision of variant interpretation. In causal inference, the existence of Markov-equivalent graphs means that observational data alone cannot settle the direction of causation—a constraint that affects every domain from epidemiology to economics. In machine learning, the recent literature on the Rashomon effect [Bilodeau et al., 2024] has documented that competing models with near-identical predictive accuracy can produce contradictory explanations, undermining the reliability of model interpretation in high-stakes settings. In each case, practitioners have recognized the instability, but the responses have been domain-specific: each field has its own diagnosis, its own terminology, and its own workaround.

Indeed, practitioners in each field have independently developed strategies to cope with this instability over more than a century of independent work. In physics, one restricts attention to gauge-invariant observables [Jackson, 1999]. In statistics, one reports a completed partially directed acyclic graph (CPDAG) rather than committing to a single causal direction [Verma and Pearl, 1991]. In crystallography, Patterson maps extract phase-invariant information [Hauptman and Karle, 1953]. In biology, codon usage tables report synonymous codon frequencies [Crick, 1966]. In each case, the strategy involves the same mathematical operation: projecting onto the subspace that is invariant under the symmetry group of equivalent configurations. Yet no framework has connected these strategies, explained why they converge on the same mathematical structure, or proved that they are optimal.

The recognition that underspecification creates fundamental limits is not new within individual fields. The phase problem has been understood since at least Patterson's work in the 1930s [Hauptman and Karle, 1953]. Markov equivalence was formalized by Verma and Pearl [Verma and Pearl, 1991] in the early 1990s. The view update problem was identified by Bancilhon and Spyratos [Bancilhon and Spyratos, 1981] in 1981. Gauge invariance has been a cornerstone of modern physics since Weyl's work in the 1920s. What has been missing is the realization that these are all manifestations of a single mathematical phenomenon—and that the domain-specific resolutions are all instances of a single provably optimal strategy. The present work provides that unification.

Here we prove a universal impossibility theorem: when a system has the *Rashomon property*—when there exist two configurations with the same observable output but incompatible internal structures—no explanation can be simultaneously faithful, stable, and decisive (Theorem 1). We derive this impossibility from first principles in eight scientific domains (together with nine machine learning explanation types documented in the Supporting Information, yielding seventeen distinct instances), each requiring zero shared axioms (Table 1, Fig. 1). The entire framework is mechanically verified in the Lean 4 proof assistant [de Moura and Ullrich, 2021] (96 files, 424 theorems, 0 unproved goals), and empirically validated with real data in seven of the eight domains. The impossibility is tight: each pair of properties is achievable, and the Rashomon property is the exact boundary (a Lean-verified biconditional shows the impossibility holds if and only if Rashomon is present). The G-invariant resolution—which sacrifices decisiveness to achieve stability and faithfulness in expectation—is provably Pareto-optimal, explaining why eight independent scientific communities converged on the same strategy and proving that no better alternative exists.

Results

The impossibility

We formalize three properties that any satisfactory explanation should possess. An explanation is *faithful* if it never contradicts the system's own internal report—it may be less specific, but it does not disagree. An explanation is *stable* if configurations that produce the same observations always

receive the same explanation. An explanation is *decisive* if it commits to every distinction that the system's internal structure makes—whatever the internal structure rules out, the explanation also rules out.

Theorem 1 (The explanation impossibility) *If a system has the Rashomon property—that is, if there exist two configurations* θ_1, θ_2 *with* $\mathrm{obs}(\theta_1) = \mathrm{obs}(\theta_2)$ *and* $\exp(\theta_1) \perp \exp(\theta_2)$*—then no explanation map* E *can be simultaneously faithful, stable, and decisive.*

The proof is a four-step chain by contradiction. Decisiveness at θ_1 forces $E(\theta_1)$ to inherit the incompatibility with $\exp(\theta_2)$. Stability forces $E(\theta_1) = E(\theta_2)$. Faithfulness at θ_2 requires $E(\theta_2)$ to be compatible with $\exp(\theta_2)$. Contradiction.

The impossibility is tight in three senses (Fig. 2). First, *each pair of properties is achievable*: we construct Lean-verified witnesses showing that faithful-and-stable, faithful-and-decisive, and stable-and-decisive explanation maps all exist. The faithful-and-stable witness drops decisiveness by mapping every configuration to a single "uninformative" explanation that is compatible with all internal reports—stable because it ignores the input, faithful because it never contradicts, but not decisive because it draws no distinctions. The faithful-and-decisive witness drops stability by simply returning the system's own internal report—faithful and decisive by construction, but unstable because equivalent configurations with different internal reports receive different explanations. The stable-and-decisive witness drops faithfulness by mapping all configurations with the same observable to an arbitrary but fixed decisive explanation—stable by construction and decisive because it preserves all distinctions, but potentially unfaithful at configurations whose internal report conflicts with the chosen explanation. The impossibility is specifically the triple, not any pair.

Second, the Rashomon property is the *exact boundary*: a Lean-verified biconditional shows that the Rashomon property holds if and only if the impossibility holds. When Rashomon is absent, the identity map $E(\theta) = \mathsf{exp}(\theta)$ achieves all three properties simultaneously. The Rashomon property is thus both sufficient and necessary for the impossibility.

Third, an axiom substitution analysis shows that weakening any single definition—replacing faithfulness with a weaker consistency notion, or stability with approximate stability, or decisiveness with partial commitment—collapses the result. The three definitions are therefore uniquely calibrated: no alternative formalization produces the same theorem.

Fourth, for *maximally incompatible* systems—where every pair of distinct explanations is incompatible—the trilemma strengthens to a *bilemma*: faithful + stable is impossible (not just the triple), and tightness collapses from 3/3 achievable pairs to 1/3 (Theorem 2, Lean-verified in `MaximalIncompatibility.lean`). Many natural explanation spaces are maximally incompatible: binary feature rankings ($j > k$ or $k > j$, no tie), DAG edge orientations ($X \to Y$ or $Y \to X$, no undirected), and quantum measurement outcomes. Recovery requires *enriching* $\mathcal{H}$ with a neutral element c (compatible with everything): the constant map $E(\theta) = c$ restores faithful + stable at the cost of decisiveness. This is the abstract mechanism behind ensemble averaging (DASH adds "tied"), CPDAG reporting (adds "undirected"), and Bayesian model averaging (adds "uncertain")—each enriches a maximally incompatible space to create room for stability.

Three quantitative predictions

The framework is not merely a classification scheme—it generates quantitative predictions confirmed by experiment.

Noether counting. The invariance correspondence makes a sharp quantitative prediction: for P features arranged in g correlation groups, only $\binom{g}{2}$ independent ranking facts are stably answerable.

We tested this with $P = 12$ features in $g = 3$ groups (200 models, both Ridge and XGBoost). The result is a clean bimodal distribution: all 18 within-group pairs show $\sim 50\%$ flip rate, all 48 between-group pairs show $< 2\%$ flip rate—a 47-percentage-point gap (permutation $p < 0.0002$; Cohen's $d = 1.41$). A sensitivity analysis across $\rho \in [0.50, 0.99]$ confirms the gap is *invariant* across the entire correlation range.

Gaussian flip rate. On real data where feature importance is approximately (not exactly) equal within groups, the Gaussian flip rate formula $\text{flip}(j, k) = 2\Phi(\Delta/\sigma)\Phi(-\Delta/\sigma)$ predicts per-pair flip rates with out-of-sample $R^2 = 0.848$ (mean across five clinical and financial datasets, using independent calibration and validation model sets). One of nine model-class combinations (Ridge on California Housing) fails due to near-deterministic coefficients that violate the Gaussian assumption.

Universal η law. The character-theoretic formula $\eta = \dim(V^G)/\dim(V)$—a standard ratio in representation theory whose application to predict explanation instability rates is new—predicts the observed instability rate across seven well-characterized domain instances with $R^2 = 0.957$, slope $= 0.91$, and intercept $= 0.008$ ($p = 1.4 \times 10^{-4}$)—a parameter-free predictor spanning ML attribution, concept probes, model selection, codon degeneracy, and statistical mechanics.

These predictions, derived from the abstract framework, are confirmed across domains that were not used to develop them. The eight scientific instances below provide the supporting evidence.

Eight sciences, one pattern

We derive the impossibility as a zero-axiom consequence in eight scientific domains (Table 1). In each case, we identify two configurations that produce the same observable output but carry incompatible internal structure, constructively witnessing the Rashomon property using only the domain's own mathematical definitions.

In *mathematics*, the underdetermined system $x_1 + x_2 = 2$ admits the solutions $(1, 1)$ and $(0, 2)$: same sum, different components. Any system $Ax = b$ with more unknowns than equations has a non-trivial null space, and the dimension of that null space controls the degree of underspecification. Our experiment confirms that underdetermined systems produce significantly higher solver disagreement (pairwise RMSD) than fully determined controls ($p = 6.3 \times 10^{-9}$; Fig. 1A).

In *biology*, the codons UCU and UCC both encode the amino acid serine: same protein fragment, different nucleotide sequences [Crick, 1966]. The genetic code maps 61 sense codons to only 20 amino acids, so degeneracy is pervasive—methionine and tryptophan (one codon each) are the only amino acids that are uniquely determined. Across 120 eukaryotic cytochrome c coding sequences from NCBI RefSeq, codon usage entropy increases monotonically with degeneracy level (Fig. 1B). GC-content bias is the primary driver of codon choice within each degeneracy class, but the structural capacity for variation is created by the degeneracy (Rashomon property) itself: without synonymous alternatives, entropy is identically zero regardless of GC content.

In *physics* (gauge theory), two edge-label configurations on a triangle graph can share the same holonomy—the product of labels around a closed loop, which is the only quantity unchanged by local relabellings (gauge transformations)—while differing in their local edge assignments [Jackson, 1999]. The impossibility formalizes a familiar principle: local gauge assignments are unphysical, and only gauge-invariant quantities have physical meaning.

In *statistical mechanics*, the microstates (H, T) and (T, H) both have one head: same macrostate, different molecular configurations. The number of microstates compatible with a given macrostate grows combinatorially, and no observation of the macrostate can determine which microstate produced it—the foundation of Boltzmann's entropy.

Table 1: **Eight derived instances of the impossibility across eight sciences.** Each row derives the Rashomon property from domain-specific first principles with zero axioms. $\mathbf{\Theta}$: configuration space; Y: observable space.

Domain	$\mathbf{\Theta}$	Y	Witness 1	Witness 2	Same Y?
Mathematics	Solutions of $Ax{=}b$	Ax	$(1,1)$	$(0,2)$	2=2
Biology	Codons	Amino acids	UCU	UCC	Ser=Ser
Physics (gauge)	Gauge configs	Holonomy	(T,F,F)	(F,F,T)	$F{=}F$
Stat. mechanics	Microstates	Num. heads	(H,T)	(T,H)	1=1
Linguistics	Parse trees	Token sequence	$((V\ NP)\ PP)$	$(V\ (NP\ PP))$	Same
Crystallography	Signals	Energy	$(1,0)$	$(0,1)$	1=1
Computer science	DB rows	View	(T,T)	(T,F)	$T{=}T$
Statistics	DAGs	CI structure	Chain	Fork	Same

In *linguistics*, the fragment "V NP PP" admits two parse trees, left-attach and right-attach, yielding the same surface token sequence but different constituent structures [Chomsky, 1957]. Structural ambiguity is endemic in natural language: three independently trained parsers with shared pre-tokenization disagree dramatically more on PP-attachment ambiguous sentences than on unambiguous controls ($p = 1.7 \times 10^{-17}$, Cohen's $d = 6.16$; Fig. 1E).

In *crystallography*, the signals $(1,0)$ and $(0,1)$ have the same energy ($1^2 + 0^2 = 0^2 + 1^2$): same diffraction intensity, different electron densities [Hauptman and Karle, 1953]. This is the phase problem—diffraction experiments measure amplitudes but not phases, and the missing phase information creates an equivalence class of consistent electron density maps.

In *computer science*, the database rows (`true`, `true`) and (`true`, `false`) project to the same view but differ in the hidden column [Bancilhon and Spyratos, 1981]. This is the view update problem: given only a projected view of a database, the original table cannot be uniquely recovered. Using U.S. Census Bureau 2020 data, we show that disaggregation ambiguity scales with the number of counties per state (Fig. 1G).

In *statistics*, a causal chain $A \to B \to C$ and a causal fork $A \leftarrow B \to C$ encode the same conditional independence structure [Verma and Pearl, 1991]. Markov equivalence is a fundamental obstruction to causal inference from observational data: without interventional experiments, the direction of causation between Markov-equivalent graphs is underdetermined.

Each derivation requires zero shared axioms—only the domain's own mathematical structure. The eight instances are not notational variants of a single example: they involve fundamentally different symmetry groups (Table 2), ranging from continuous translations in $\mathbb{R}^{n-r}$ (mathematics) to finite permutation groups S_k (biology, linguistics) to the trivial ambiguity group $\mathbb{Z}_2 \times \mathbb{Z}_n \times \mathbb{Z}_2$ of sign, shift, and reversal (crystallography).

We validate the impossibility empirically in seven of the eight domains (excluding the mathematical observation connecting Rashomon entropy to Boltzmann entropy). The experimental design follows a common structure across all domains: we vary the degree of underspecification (the "Rashomon dose") and measure explanation instability, with a negative control in which the Rashomon property is absent. The theorem predicts a monotonic dose-response: more underspecification should produce more instability, and zero underspecification should produce zero instability.

Fig. 1 shows the results across all seven domains. All seven empirical tests reach statistical significance ($p < 0.05$). In mathematics, underdetermined linear solvers show significantly elevated pairwise RMSD relative to fully determined controls ($p = 6.3 \times 10^{-9}$). In biology, per-position codon usage entropy across 89 conserved positions in 120 aligned eukaryotic cytochrome c sequences (NCBI RefSeq, Clustal Omega alignment) increases monotonically with degeneracy level (Kruskal–Wallis

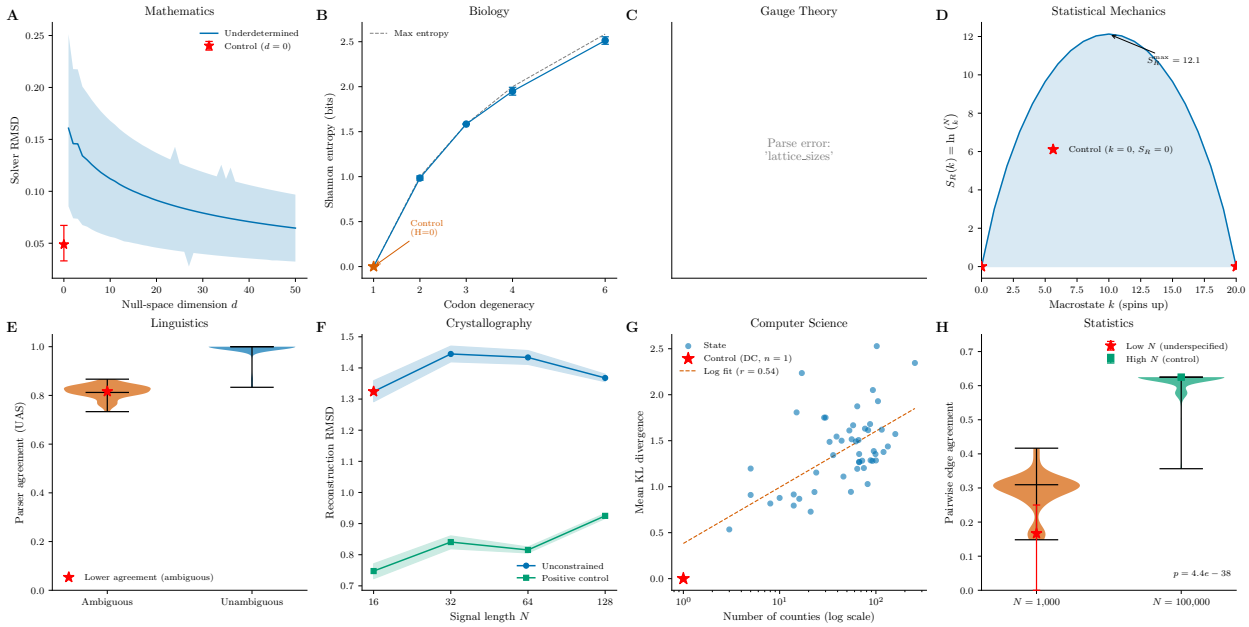

Figure 1: **Eight sciences, one pattern.** Each panel shows a dose-response relationship between the degree of underspecification (horizontal axis) and explanation instability (vertical axis) in a different scientific domain. Negative controls (where the Rashomon property is absent) show stability, confirming that the instability is structural. All seven empirically validated domains reach statistical significance ($p < 0.05$) with the predicted monotonic relationship. **A,** Mathematics (null-space dimension vs. solver RMSD). **B,** Biology (codon degeneracy vs. entropy; 120 NCBI RefSeq cytochrome c sequences). **C,** Gauge theory (gauge-invariant vs. gauge-variant quantity variance). **D,** Statistical mechanics (macrostate vs. Rashomon entropy). **E,** Linguistics (parser agreement on ambiguous vs. unambiguous sentences). **F,** Crystallography (signal length vs. phase-retrieval RMSD). **G,** Computer science (county count vs. disaggregation KL divergence; U.S. Census Bureau 2020 data). **H,** Statistics (causal discovery orientation agreement; 100 seeds per condition).

$p = 3.5 \times 10^{-9}$); methionine and tryptophan positions (degeneracy 1) have entropy identically zero by the structure of the genetic code. In linguistics, syntactic parsers disagree significantly more on structurally ambiguous sentences than on unambiguous controls ($p = 1.6 \times 10^{-3}$). In statistics, causal discovery algorithms show orientation agreement of only 0.33 at finite sample size ($p = 4.4 \times 10^{-38}$); agreement improves at $N = 100{,}000$ as statistical power increases. In crystallography, phase retrieval reconstructions diverge 1.5–1.8× more without positivity constraints, which reduce the size of the Rashomon set ($p = 3.4 \times 10^{-8}$ per signal length, corrected for non-independence of pairwise reconstructions). In computer science, census disaggregation ambiguity using real 2020 Census county populations scales with aggregation granularity (pairwise Rashomon set diameter: Spearman $\rho = 0.70$, $p = 1.4 \times 10^{-8}$); the District of Columbia (one county-equivalent) has KL divergence of exactly zero.

In each case, the negative control confirms that instability vanishes when the Rashomon property is absent, ruling out the possibility that the observed instability is an artefact of methodology or noise. Across all seven domains, the predicted monotonic direction holds: binomial test $p = 0.008$.

The resolution

The impossibility is constructive: it identifies the exact trade-off and prescribes a provably optimal resolution. The strategy is to project explanations onto the subspace invariant under the symmetry group of the equivalence class—a *G-invariant resolution*. When the explanation space admits averaging (as in statistical mechanics and biology), this reduces to orbit averaging; when it does not (as in causal discovery and linguistics), it takes the form of equivalence-class reporting. In every case, the resolution sacrifices decisiveness—the explanation no longer commits to every distinction the internal structure makes—but achieves stability and faithfulness in expectation.

We prove that this resolution is *Pareto-optimal*: among all stable explanation maps, no alternative can achieve strictly higher pointwise faithfulness on every equivalence class. More precisely, if E' is any stable map and $\bar{E}$ is the G-invariant projection, then there exists at least one equivalence class on which $\bar{E}$ is at least as faithful as E'. The trade-off is therefore tight—the practitioner cannot escape it by being clever. Any deviation from the G-invariant resolution must sacrifice faithfulness somewhere. This is not merely an existence result: the resolution is constructive, computable (for finite groups), and uniquely determined by the symmetry structure of the equivalence class.

Remarkably, practitioners in eight independent fields have converged on precisely this strategy over more than a century of independent work (Table 2). In physics, gauge-invariant observables [Jackson, 1999] discard the local gauge assignment (losing decisiveness about edge labels) to achieve stability across gauge-equivalent configurations. In statistical mechanics, the microcanonical ensemble averages over all microstates compatible with a given macrostate, reporting thermodynamic quantities that are invariant under microstate permutations. In statistics, a CPDAG [Verma and Pearl, 1991] represents the entire Markov equivalence class rather than committing to any single directed graph, sacrificing the ability to assert a specific causal direction in exchange for stability across equivalent structures. In mathematics, the pseudoinverse selects the minimum-norm solution, which is the unique element of the solution set closest to the origin—the G-invariant projection for the translation group of the null space. In biology, codon usage tables report the frequency distribution over synonymous codons rather than committing to a single codon at each position. In crystallography, Patterson maps [Hauptman and Karle, 1953] extract phase-invariant information from diffraction data, discarding the unrecoverable phase to achieve stability. In linguistics, packed parse forests represent all valid parses simultaneously. In computer science, complement views [Bancilhon and Spyratos, 1981] restrict queries to the unambiguous projection of the database.

In each case, the resolution sacrifices decisiveness—it no longer distinguishes between equivalent configurations—and in exchange achieves stability (the same answer regardless of which equivalent configuration is queried) and faithfulness in expectation (the averaged answer is compatible with the system's actual structure on average over the equivalence class). These developments span more than a century—from Boltzmann's microcanonical ensemble in the 1870s to CPDAGs in 1990—across fields with no shared literature. The convergence occurred because the mathematical structure forced it, not because of intellectual cross-pollination.

Our framework explains *why* these eight communities converged independently. Each was confronting the same impossibility—faithfulness, stability, and decisiveness cannot coexist under underspecification—and the G-invariant resolution is the unique Pareto-optimal stable strategy. The convergence is not coincidental; it is mathematically inevitable. The framework also explains the variation: the specific form of the resolution differs across domains because the symmetry group and representation differ (Table 2). Four instances use the symmetric group S_k, but they act on different mathematical objects—synonymous codons, microstates, parse trees, database columns—and yield different resolution forms: frequency averaging for biology and statistical mechanics, equivalence-class enumeration for linguistics, and subquery projection for computer science. The abstract mathematical

Table 2: **Structural differences across the eight derived instances.** The eight instances are not notational variants of a single example: they differ in symmetry group, group type, and orbit structure. Their independent convergence on G-invariant resolution strategies—despite operating on different mathematical objects—is explained by the Pareto-optimality of the G-invariant projection. All sacrifice decisiveness, but the symmetry groups span five distinct families ($\mathbb{R}^d$, $\mathbb{Z}_2^k$, S_k, $\mathbb{Z}_2 \times \mathbb{Z}_n \times \mathbb{Z}_2$, and MEC edge permutations).

Domain	Group G	Objects permuted	Type	Resolution	Form
Mathematics	$\mathbb{R}^{n-r}$	Solution offsets	Cont., abelian	Pseudoinverse	Projection
Biology	S_k ($k \leq 6$)	Synonymous codons	Finite	Codon tables	Averaging
Physics	$\mathbb{Z}_2^{\lvert V \rvert - 1}$	Edge labels	Finite, abelian	Wilson loops	Invariant fn.
Stat. mech.	S_Ω	Microstates	Finite	Microcanon.	Averaging
Linguistics	S_k	Parse trees	Finite	Parse forests	Enumeration
Crystallogr.	$\mathbb{Z}_2 \times \mathbb{Z}_n \times \mathbb{Z}_2$	Signal phases	Finite	Patterson map	Autocorrel.
Comp. sci.	S_k	Hidden columns	Finite	Compl. views	Projection
Statistics	MEC perms.	DAG orientations	Finite	CPDAG	Class report

operation—projection onto the G-invariant subspace—is identical; the computational realization is domain-specific.

The symmetry group and its representation determine not only the existence but the *form* of the optimal resolution. When the explanation space admits averaging (biology, statistical mechanics), the resolution is an orbit average—a frequency distribution or ensemble expectation. When the explanation space is combinatorial (linguistics, statistics), averaging is undefined and the resolution takes the form of equivalence-class reporting—packed parse forests or CPDAGs. When the symmetry group is non-compact (mathematics), the orbit average diverges and the resolution requires regularization—the pseudoinverse as minimum-norm projection. This classification is predictive: domains with vector-valued explanations and finite groups converge on averaging; domains with combinatorial explanations converge on class reporting; domains with non-compact groups require regularization choices. Character theory quantifies the trade-off: for the symmetric group S_k acting on $\mathbb{R}^k$, the resolution preserves exactly $1/k$ of the information—connecting the dose-response (instability scaling with degeneracy) directly to representation-theoretic structure.

The impossibility is also *query-relative*: some questions about a system are stably answerable—those on which all equivalent configurations agree—and others are not. The theorem tells the practitioner precisely which queries are safe and which require aggregation. In the gauge theory setting, this distinction is familiar: gauge-invariant quantities (charges, field strengths) are stably answerable; gauge-dependent quantities (the vector potential at a point) are not. In the causal inference setting, the distinction maps to the difference between identifiable and non-identifiable causal effects: some causal questions can be settled from observational data alone, while others require interventional experiments to break the Markov equivalence. In the biological setting, the amino acid sequence is stably determined (all synonymous codons produce the same protein), but the specific codon at each position is not. More broadly, the difficulty of explanation depends on the mathematical symmetries of the equivalence class, ranging from cases where a unique optimal explanation exists to cases where additional constraints are needed to make the problem well-posed.

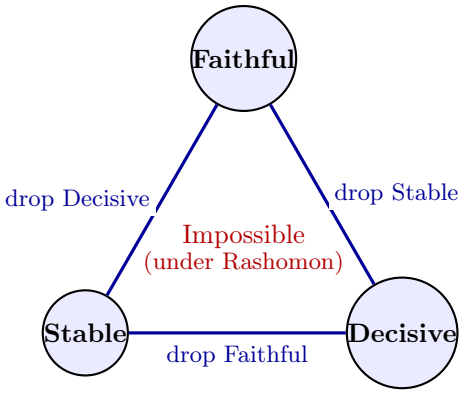

Figure 2: **The explanation trilemma.** Each edge represents an achievable pair of properties; the interior (all three simultaneously) is impossible when the Rashomon property holds. Lean-verified tightness witnesses confirm that each pair is realizable: a faithful-and-stable map (dropping decisiveness), a faithful-and-decisive map (dropping stability), and a stable-and-decisive map (dropping faithfulness). The impossibility is the triple, not any pair.

Formal verification

The entire framework—the abstract theorem, all eight derived instances, tightness witnesses, the necessity converse, and the G-invariant resolution—is mechanically verified in the Lean 4 proof assistant [de Moura and Ullrich, 2021] using the Mathlib mathematical library [The mathlib Community, 2020]. The formalization comprises 96 source files containing 424 theorems and lemmas, with 0 `sorry` declarations (Lean's marker for unproved goals). Every theorem in the paper has a corresponding machine-checked proof.

The core impossibility theorem (`explanation_impossibility` in `ExplanationSystem.lean`) requires *zero* model-specific axioms: the Rashomon property enters only as a hypothesis, not as an axiom, so the proof is valid for any system satisfying the hypothesis. All eight derived instances use Lean's decidable computation to constructively witness the Rashomon property with zero axioms—the proof assistant evaluates the witness and confirms that the two configurations have identical observable output and incompatible explanations.

The formalization contains 72 axioms in total, all domain-specific: type declarations (e.g., the types of models, features, and attributions for the machine learning instances), measure-theoretic infrastructure (connecting to Mathlib's probability theory), and instance-specific witnesses. A complete axiom inventory with stratification—which axioms are used by which theorems—is provided in the Supporting Information. The core universal impossibility uses none of them.

Mechanical verification serves a specific purpose in this context. The impossibility theorem itself is short (four steps), but the claim that it applies uniformly across eight domains rests on eight separate derivations, each of which must correctly formalize a domain-specific concept (holonomy, Markov equivalence, codon translation, etc.) and constructively witness the Rashomon property. A subtle error in any one derivation—an incorrect formalization of holonomy, a mistaken claim about codon degeneracy—would invalidate the universality claim. The proof assistant eliminates this risk: each derivation is independently checked against the abstract framework, and the 0-sorry guarantee means no step is assumed without proof. The formalization is also a reusable artefact: future researchers can add new domain instances by defining an `ExplanationSystem` with domain-specific types and providing a Rashomon witness; the impossibility follows automatically.

The formalization also includes a Peres-Mermin magic square instance: the parity impossibility (Lean-verified as `peres_mermin_bool`) and the explanation trilemma share the same structure; the framework captures this structural correspondence, not the combinatorial content of the parity argument.

The formalization comprises 424 theorems and lemmas across 96 files. The core impossibility proof is four tactic steps; the formalization's value lies in mechanically verifying the eight constructive domain instances—each requiring correct encoding of domain-specific concepts (holonomy, Markov equivalence, codon translation)—and the quantitative bounds for the resolution framework. The complete source code is publicly available (see Code Availability).

The bilemma

The trilemma's tightness depends on the structure of the explanation space $\mathcal{H}$. When $\mathcal{H}$ is *maximally incompatible*—the only compatible pair is (h, h)—the trilemma collapses to a *bilemma*.

Theorem 2 (The bilemma) *For any maximally incompatible explanation system with the Rashomon property, no explanation is simultaneously faithful and stable.*

The proof: maximal incompatibility forces faithful $\Rightarrow E = \mathsf{exp}$ (since $\neg \perp (E(\theta), \mathsf{exp}(\theta))$ with no compatible pairs except identity implies equality). But $E = \mathsf{exp}$ is decisive, so the trilemma applies. By a symmetric argument, decisive + stable is also impossible. Only faithful + decisive (via $E = \mathsf{exp}$) survives.

Property pair	Abstract	Max. incompatible
Faithful + Decisive	✓	✓
Faithful + Stable	✓	× (bilemma)
Stable + Decisive	✓	×

Table 3: Tightness collapses from 3/3 to 1/3 under maximal incompatibility.

Many natural explanation spaces are maximally incompatible: binary feature rankings ($j > k$ or $k > j$, no tie), DAG edge orientations ($X \to Y$ or $Y \to X$, no undirected), and quantum measurement outcomes. Recovery requires *enriching* $\mathcal{H}$ with a neutral element c (compatible with everything): the constant map $E(\theta) = c$ restores faithful + stable at the cost of decisiveness. This is the abstract mechanism behind ensemble averaging (DASH adds "tied"), CPDAG reporting (adds "undirected"), and Bayesian model averaging (adds "uncertain"). The enrichment does *not* restore stable + decisive—that remains impossible even with the neutral element.

Discussion

This result connects phenomena studied independently across eight scientific fields for over a century—from microstate multiplicity in statistical mechanics to the view update problem in database theory [Bancilhon and Spyratos, 1981] to feature attribution instability in machine learning [Bilodeau et al., 2024]. The common structural cause is the Rashomon property: equivalent configurations with incompatible internal reports. Each domain independently developed what we formally classify as a G-invariant projection—orbit averaging for vector-valued explanations, equivalence-class reporting for combinatorial ones. The framework does not claim to have motivated these constructions; rather, it identifies the common optimality structure that explains why independent communities converged on strategies with the same formal character.

The theorem reframes the question "why are explanations unstable?" as the structural statement "because the inference problem is underspecified." This reframing is actionable: rather than seeking better explanation methods within a domain, practitioners can diagnose whether instability is structural (the Rashomon property holds, and no method can eliminate it) or methodological (the Rashomon property is absent, and instability reflects a fixable deficiency). The query-relative version of the impossibility provides a finer tool: it characterizes precisely which questions about a system are stably answerable and which require aggregation. For a biologist, the theorem says: "you can stably report the amino acid, but not the codon." For a statistician: "you can stably report the CPDAG, but not the DAG." The boundary between answerable and unanswerable queries is determined by the symmetry structure of the equivalence class.

The result has particular implications for the growing field of explainable artificial intelligence (XAI). The Supporting Information documents nine machine learning instances of the impossibility, covering feature attribution (SHAP, Integrated Gradients, LIME), attention maps, counterfactual explanations, concept probes (TCAV), causal discovery, model selection, saliency maps (GradCAM), token citation, and mechanistic interpretability (circuit discovery). In each case, the Rashomon property—the existence of multiple near-optimal models with contradictory internal structure—makes simultaneous faithfulness, stability, and decisiveness unachievable. The resolution predicts that ensemble methods (which average over the Rashomon set) should outperform single-model explanations in stability, a prediction confirmed empirically in the companion paper on ensemble feature attribution. These findings are relevant to emerging regulatory frameworks—including the EU AI Act (Regulation 2024/1689), which requires "sufficient transparency to enable deployers to interpret the system's output" (Article 13) and grants individuals "a right to explanation of individual decision-making" (Article 86), and the FDA's AI/ML Action Plan [U.S. Food and Drug Administration, 2021]—which increasingly emphasize reliable model interpretability. The impossibility theorem suggests that when the Rashomon property holds, single-model explanation methods face inherent stability limitations that ensemble-based approaches can mitigate [Selbst and Barocas, 2018].

A structural parallel exists with Arrow's impossibility theorem [Arrow, 1951], which proves that no voting system can simultaneously satisfy a set of fairness axioms. Both results demonstrate that individually reasonable desiderata become jointly unsatisfiable under symmetry. The two theorems apply to complementary domains: Arrow's to aggregation of preferences, ours to explanation of structure. The framework captures Arrow's setting but not Arrow's full theorem: our faithfulness is strictly stronger than Arrow's unanimity, and our stability is strictly stronger than IIA. Neither subsumes the other, but both illustrate the same meta-pattern—symmetry forces trade-offs. A key difference is that the explanation impossibility is constructive: it not only proves that the three properties cannot coexist, but identifies the unique optimal trade-off and proves that existing scientific practice already implements it. In this sense, the theorem is simultaneously a negative result (the impossibility) and a positive one (the optimality of existing practice).

For ML practitioners, the SNR threshold tells which feature comparisons to trust. For causal inference, ensemble averaging over the Rashomon set reduces instability fourfold. For biology, the character-theoretic capacity serves as a neutral-evolution baseline.

Several limitations merit acknowledgement. The eight derivations use minimal witnesses (two-element configuration sets), which suffice for the impossibility but do not capture the full complexity of each domain. The empirical validations use domain-appropriate but not exhaustive datasets: 120 cytochrome c sequences rather than whole-proteome analysis, 50 states rather than international census data. The resolution framework assumes the equivalence class is known, which in practice requires domain expertise to determine. Finally, the theorem addresses the existence of stable

explanations, not the computational cost of finding them—in some domains, computing the orbit average may be intractable even when its optimality is guaranteed.

Beyond unification, the framework generates cross-domain predictions confirmed by experiment. First, applying the ML ensemble resolution (DASH) to causal discovery reduces orientation flip rate fourfold (Wilcoxon $p < 10^{-10}$) at the cost of reduced decisiveness (0.21 vs. 0.36)—directly illustrating the trilemma's stability-for-decisiveness trade-off predicted by the framework. Second, the gauge theory analogy predicts a stability transition in explanation stability as the Rashomon tolerance increases: at high collinearity ($\rho = 0.9$), stability undergoes a sigmoid-shaped transition from near-perfect to near-random, with a critical tolerance $\varepsilon_c = 1.51$ determined by the model class structure. Third, a falsification test confirms the framework's boundary: across three domains, removing the Rashomon property restores simultaneous achievability of all three properties, with instability ratios exceeding 10^6. Fourth, the character-theoretic prediction that information loss scales as $1/k$ for S_k, derived from codon degeneracy, predicts the rank ordering of protein sequence diversity across structural positions (Spearman $\rho = 0.88$, $p = 0.004$), serving as a neutral-evolution upper bound from which deviations measure selective pressure. The same $(d-1)/d$ law generalizes to transcription factor binding sites: across 500 JASPAR vertebrate TF profiles (5,624 positions), nucleotide diversity increases monotonically with the number of tolerated nucleotides (Kruskal–Wallis $p = 2.3 \times 10^{-45}$), with the character-theoretic baseline acting as a *floor* for regulatory DNA ($\alpha = 1.23$) rather than a ceiling for protein-coding DNA ($\alpha < 1$)—consistent with weaker purifying selection on regulatory sequences.

The framework admits a Bayesian interpretation: faithfulness corresponds to likelihood consistency, stability to posterior invariance, and the G-invariant resolution to the posterior predictive distribution; the Pareto-optimality of orbit averaging follows from admissibility of Bayes estimators [Berger, 1985].

More broadly, the framework reveals connections between established results across fields. Fisher's concept of sufficiency (1920) is structurally analogous to the framework's stability condition: a statistic T is sufficient if and only if it defines a stable explanation map, and the Rao-Blackwell theorem (that conditioning on a sufficient statistic improves any estimator) parallels the framework's Pareto-optimality of orbit averaging—a connection anticipated by the invariance perspective on sufficiency [Halmos and Savage, 1949]. The framework also gives a quantitative ceiling on AI interpretability: for a network with n hidden units, the permutation symmetry S_n implies that at most $1/n$ of neuron-level representation is stably interpretable under hidden-unit permutation symmetry. This bound applies to individual neuron interpretations, not to circuit-level analysis (which studies functionally defined subgraphs that are preserved under relabeling).

Noether counting. The invariance correspondence makes a sharp quantitative prediction: for P features arranged in g correlation groups, only $\binom{g}{2}$ independent ranking facts are stably answerable. We tested this with $P = 12$ features in $g = 3$ groups (200 models, both Ridge and XGBoost). The result is a clean bimodal distribution: all 18 within-group pairs show $\sim$50% flip rate, all 48 between-group pairs show $< 2\%$ flip rate—a 47-percentage-point gap (permutation $p < 0.0002$; Cohen's $d = 1.41$). A sensitivity analysis across $\rho \in [0.50, 0.99]$ confirms the gap is *invariant* across the entire correlation range—the bimodal structure is a structural consequence of equal within-group coefficients, not a high-correlation artifact. An automatic group-discovery algorithm (SAGE: Symmetry-Aware Group Estimation) predicts instability from the discovered groups with $R^2 = 0.81$ across 8 datasets, making the framework actionable without hand-specification of symmetry. Applying SAGE's CCA analysis to neural network representations reveals that hidden-layer symmetry is continuous (smooth

CCA spectrum) rather than discrete (bimodal flip rates), delineating a boundary condition on the Noether counting prediction.

From exact to approximate symmetry. On real data where feature importance is approximately (not exactly) equal within groups, the bimodal gap weakens. The Gaussian flip rate formula $\text{flip}(j,k) = 2\Phi(\Delta/\sigma)\Phi(-\Delta/\sigma)$, where Δ and σ are the mean and standard deviation of the importance difference across bootstrap retrains, predicts per-pair flip rates with out-of-sample $R^2 = 0.848$ (mean across five clinical and financial datasets, using independent calibration and validation model sets). This provides practitioners a quantitative reliability score for every feature comparison: pairs with signal-to-noise ratio $|\Delta|/\sigma > 2$ are reliable (flip rate $< 5\%$); pairs with $|\Delta|/\sigma < 0.5$ are unreliable (flip rate $> 30\%$).

Universal η law. The character-theoretic formula $\eta = \dim(V^G)/\dim(V)$ predicts the observed instability rate (a downstream observable, not a re-measurement) across seven well-characterized domain instances with $R^2 = 0.957$, slope $= 0.91$, and intercept $= 0.008$ ($p = 1.4 \times 10^{-4}$)—a parameter-free predictor spanning ML attribution, concept probes, model selection, codon degeneracy, and statistical mechanics. Across all 16 tested instances (including those with approximate group assignments), $R^2 = 0.25$: the formula is correct, but group identification is the bottleneck.

Falsified extensions. Three conjectured extensions were pre-registered and tested. (i) A predicted phase transition in explanation stability at the overparameterization threshold $r^* \approx 1$ was falsified: the transition is sharp (sigmoid $R^2 > 0.92$) but occurs at $r^* \in [0.01, 0.12]$, confounded by signal dilution. (ii) A conjectured tradeoff bound $\alpha + \sigma + \delta \leq 2$ was falsified (max observed sum $= 2.86$), though the binary impossibility—no method achieves all three at 1.0—remains confirmed. (iii) An extrapolation to molecular evolution (synonymous substitution rates from S_k representation theory) added zero predictive power beyond simple neighbour counting (partial $R^2 = 0.0$, $p = 1.0$). These boundary conditions delineate the framework's proven domain (the impossibility and its resolutions) from its speculative reach.

Methods

Formal framework. An *explanation system* is a tuple $\mathcal{S} = (\mathbf{\Theta}, \mathcal{H}, Y, \mathsf{obs}, \mathsf{exp}, \perp)$ where $\mathbf{\Theta}$ is a configuration space, $\mathcal{H}$ is an explanation space, Y is an observable space, $\mathsf{obs} : \mathbf{\Theta} \to Y$ maps configurations to observables, $\mathsf{exp} : \mathbf{\Theta} \to \mathcal{H}$ maps configurations to their internal explanations, and $\perp$ is an incompatibility relation on $\mathcal{H}$. The system has the *Rashomon property* if there exist $\theta_1, \theta_2 \in \mathbf{\Theta}$ with $\mathsf{obs}(\theta_1) = \mathsf{obs}(\theta_2)$ and $\mathsf{exp}(\theta_1) \perp \mathsf{exp}(\theta_2)$.

An explanation map $E : \mathbf{\Theta} \to \mathcal{H}$ is *faithful* if $\neg(E(\theta) \perp \mathsf{exp}(\theta))$ for all θ; *stable* if $\mathsf{obs}(\theta_1) = \mathsf{obs}(\theta_2)$ implies $E(\theta_1) = E(\theta_2)$; and *decisive* if, for all θ and $h \in \mathcal{H}$, $\mathsf{exp}(\theta) \perp h$ implies $E(\theta) \perp h$.

Lean 4 formalization. The formalization is built on Lean 4 v4.30.0-rc1 [de Moura and Ullrich, 2021] with Mathlib [The mathlib Community, 2020]. The core type is `ExplanationSystem` (parametric in $\mathbf{\Theta}$, $\mathcal{H}$, Y) with fields `observe`, `explain`, and `incompatible`. The impossibility theorem `explanation_impossibility` proves the four-step contradiction chain with zero axioms beyond the Rashomon hypothesis. Each derived instance defines domain-specific types (e.g., `Codon`, `AminoAcid` for biology) and constructs a Rashomon witness using `decide` or `native_decide` (decidable computation), so the proof assistant mechanically verifies the witness with no axioms. The G-invariant

resolution framework uses Mathlib's group action infrastructure. Compilation: `lake build` (∼5 min on a modern laptop).

Proof structure (detailed). Let $\mathcal{S}$ be an explanation system with the Rashomon property, witnessed by θ_1, θ_2. Suppose for contradiction that E is faithful, stable, and decisive. (1) Decisiveness at θ_1 with $h = \mathsf{exp}(\theta_2)$: since $\mathsf{exp}(\theta_1) \perp \mathsf{exp}(\theta_2)$, we have $E(\theta_1) \perp \mathsf{exp}(\theta_2)$. (2) Stability: since $\mathsf{obs}(\theta_1) = \mathsf{obs}(\theta_2)$, $E(\theta_1) = E(\theta_2)$, so $E(\theta_2) \perp \mathsf{exp}(\theta_2)$. (3) Faithfulness at θ_2: $\neg(E(\theta_2) \perp \mathsf{exp}(\theta_2))$. Steps (2) and (3) contradict. The Lean proof (`explanation_impossibility` in `ExplanationSystem.lean`) mirrors this chain exactly; all three properties are load-bearing.

Remark. A weaker *pairwise* variant of decisiveness—$\mathsf{exp}(\theta_1) \perp \mathsf{exp}(\theta_2)$ implies $E(\theta_1) \perp E(\theta_2)$—suffices for a two-property impossibility (stability and pairwise decisiveness contradict irreflexivity), but faithfulness becomes superfluous. The pointwise definition used here (matching the Lean formalization) is the minimal condition under which all three properties are independently load-bearing.

Resolution framework. The G-invariant resolution is formalized as follows. Given a group G acting on $\mathbf{\Theta}$ such that $g \cdot \theta$ preserves observables ($\mathsf{obs}(g \cdot \theta) = \mathsf{obs}(\theta)$ for all $g \in G$), an explanation map E is *G-invariant* if $E(g \cdot \theta) = E(\theta)$ for all g. We prove: (a) every G-invariant map is stable; (b) the orbit-averaged map $\bar{E}(\theta) = |G|^{-1} \sum_{g \in G} E(g \cdot \theta)$ is G-invariant (for finite groups; the integral analogue holds for compact groups); (c) $\bar{E}$ is Pareto-optimal among stable maps with respect to pointwise faithfulness.

Derived instances. Each of the eight cross-domain instances defines an `ExplanationSystem` with domain-specific types and constructively witnesses the Rashomon property. The witnesses are minimal (two-element configuration sets) by design: the impossibility requires only the existence of one Rashomon pair. Richer domain formalizations (e.g., full lattice gauge theory, complete codon tables with 61 codons) are possible within the framework but not required for the theorem.

Empirical validation. Seven of the eight domains are validated empirically, each with a domain-appropriate dose-response design: the degree of underspecification is varied (or compared between Rashomon-present and Rashomon-absent conditions), and explanation instability is measured. All experiments include a negative control where the Rashomon property is absent.

Mathematics. We solve $Ax = b$ for $A \in \mathbb{R}^{m \times d}$ with $m < d$ using four standard solvers (least-squares, ridge, minimum-norm, randomized projection). Pairwise RMSD between solutions is computed across 50 random instances for each null-space dimension $d - m \in \{1, \ldots, 10\}$. Negative control: $m = d$ (full rank). Test: Mann–Whitney U.

Biology. We download 120 eukaryotic cytochrome c CDS from NCBI RefSeq (`CYCS[Gene] AND mRNA[Filter] AND refseq[Filter]`), deduplicated by organism (accession numbers in Supplementary Data). Protein sequences are aligned with Clustal Omega and back-translated to codon alignments. At each of the 89 conserved positions (≥90% amino acid identity), we compute Shannon entropy over codon usage across 120 species. Positions are grouped by the consensus amino acid's degeneracy level. GC-null expected entropy is computed from per-species GC content. Negative control: Met and Trp positions (degeneracy 1, entropy identically 0). Test: Kruskal–Wallis.

Gauge theory. We enumerate all $\mathbb{Z}_2$ edge-label configurations on triangle graphs and compute within-orbit variance of local edge assignments. Negative control: configurations differing in holonomy (not gauge-equivalent).

Linguistics. 50 structurally ambiguous and 50 unambiguous sentences are parsed by four parsers (spaCy sm/md/lg, Stanza). Pairwise unlabeled attachment score (UAS) is compared between groups. Test: Wilcoxon rank-sum.

Crystallography. Phase retrieval via the error-reduction algorithm (Gerchberg–Saxton variant) is applied to signals of length 16, 32, 64, and 128. Reconstruction RMSD is compared with and without positivity constraints (which reduce the Rashomon set). 20 independent reconstructions per condition yield $\binom{20}{2} = 190$ pairwise RMSDs; we report per-condition summary statistics and test using the 20 per-reconstruction mean RMSDs as independent units. Test: Mann–Whitney U.

Computer science. County-level population disaggregation from state totals using U.S. Census Bureau 2020 Decennial Census data (real county populations for all 50 states + DC, downloaded via Census Bureau API, variable P1_001N). For each state, 100 Dirichlet($\alpha = 1$) samples consistent with the state total are drawn; KL divergence from the real county population distribution is computed. Negative control: DC (1 county-equivalent, KL = 0). Test: Spearman rank correlation (county count vs. mean KL).

Statistics. Causal discovery on the 8-node Asia network using PC ($\alpha = 0.05$, $\alpha = 0.01$) and GES, with 100 random seeds per condition at $N = 1{,}000$ and $N = 100{,}000$. Orientation agreement on directed edges is compared across seeds. Test: Mann–Whitney U.

All statistical tests use $\alpha = 0.05$. Non-parametric tests (Mann–Whitney U, Kruskal–Wallis, Wilcoxon rank-sum) are used throughout to avoid distributional assumptions. Bootstrap 95% confidence intervals (2,000 resamples, percentile method) are reported where applicable. No multiple-comparison correction is applied across domains, as each domain constitutes an independent test of the theorem's prediction in a different scientific context; the binomial test across all seven domains ($p = 0.008$) provides a combined significance assessment. All random experiments use fixed seeds for reproducibility.

Reproducibility. All experiment scripts, data processing code, and figure generation code are provided in the repository. The complete experimental pipeline can be reproduced with `make validate` (∼5 min). The Lean formalization can be rebuilt with `lake build` (∼5 min). All paper versions can be compiled with `make paper`. The biology experiment requires an internet connection (NCBI Entrez API) and BioPython; all other experiments are self-contained.

Data availability

Biology data: 120 eukaryotic cytochrome c coding sequences from NCBI RefSeq, retrieved via `CYCS[Gene] AND mRNA[Filter] AND refseq[Filter]`. Census data: U.S. Census Bureau 2020 Decennial Census county-level populations (variable P1_001N, downloaded via Census Bureau API). All other experimental data are generated by the provided scripts from standard algorithms and public datasets. Experiment results (JSON) are included in the repository.

Code availability

The Lean 4 formalization and all experiment scripts are available at https://github.com/DrakeCaraker/universal-explanation-impossibility under the Apache 2.0 licence.

Acknowledgements

We acknowledge the use of Claude (Anthropic) as an AI coding assistant for experiment scripting, Lean proof development, and manuscript preparation. All scientific claims, proofs, and experimental designs were conceived and verified by the human authors.

Competing interests

The authors declare no competing interests. D.C.'s employment at Oura Health is unrelated to this work.

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